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### Sensor

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### Abstract

According to one embodiment, a sensor includes a first sensor part. The first sensor part includes an insulating member, and first and second element parts. The first element part includes a first resistance element including first and first other resistance end portions, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member. The second element part includes a second resistance element including and second other resistance end portions, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-005387, filed on Jan. 18, 2022; the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to a sensor.

BACKGROUND

(3) For example, it is desired to improve the detection performance of the sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic plan view illustrating a sensor according to a first embodiment;

- (2) FIGS. 2A to 2F are schematic cross-sectional views illustrating the sensor according to the first embodiment;
- (3) FIGS. 3A and 3B are graphs illustrating experimental results;
- (4) FIGS. 4A and 4B are schematic views illustrating simulation results;
- (5) FIGS. 5A and 5B are schematic plan views illustrating the sensor according to the first embodiment;
- (6) FIGS. 6A and 6B are schematic cross-sectional views illustrating a sensor according to a second embodiment;
- (7) FIG. 7 is a schematic plan view illustrating a sensor according to a third embodiment;
- (8) FIG. 8 is a schematic cross-sectional view illustrating a part of the sensor according to the third embodiment;
- (9) FIG. 9 is a schematic plan view illustrating one of multiple operations in the sensor according to the third embodiment;
- (10) FIG. 10 is a schematic plan view illustrating one of multiple operations in the sensor according to the third embodiment;
- (11) FIG. 11 is a schematic plan view illustrating the sensor according to the third embodiment;
- (12) FIG. 12 is a block diagram illustrating a sensor according to the third embodiment; and
- (13) FIG. 13 is a schematic cross-sectional view illustrating a sensor according to the embodiment.

#### DETAILED DESCRIPTION

(14) According to one embodiment, a sensor includes a first sensor part. The first sensor part includes an insulating member, a first element part, and a second element part. The first element part includes a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member. The second element part includes a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member. A second distance between the second electrode and the second other electrode is shorter than a distance between the first electrode and the first other electrode.

(15) According to one embodiment, a sensor includes a first sensor part. The first sensor part includes an insulating member, a first element part, and a second element part. The first element part includes a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member. The second element part includes a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member. The insulating member includes a first insulating region between the first electrode and the first other electrode, and a second insulating region between the second electrode and the second other electrode. The insulating member includes a first material of the first insulating region, and a first contact angle of the first insulating region with respect to water. The insulating member includes at least any one of a second material of the second insulating region, the second material being different from the first material, or a second contact angle of the second insulating region with respect to water, the second contact angle being smaller than the first contact angle.

- (16) Various embodiments are described below with reference to the accompanying drawings.
- (17) The drawings are schematic and conceptual; and the relationships between the thickness and width of portions, the proportions of sizes among portions, etc., are not necessarily the same as the actual values. The dimensions and proportions may be illustrated differently among drawings, even for identical portions.
- (18) In the specification and drawings, components similar to those described previously or illustrated in an antecedent drawing are marked with like reference numerals, and a detailed description is omitted as appropriate.

#### First Embodiment

- (19) FIG. 1 is a schematic plan view illustrating a sensor according to a first embodiment.
- (20) FIGS. 2A to 2F are schematic cross-sectional views illustrating the sensor according to the first embodiment.
- (21) FIG. 2A is a cross-sectional view taken along the line A1-A2 of FIG. 1. FIG. 2B is a cross-sectional view taken along the line A3-A4 of FIG. 1. FIG. 2C is a sectional view taken along line A5-A6 of FIG. 1. FIG. 2D is a sectional view taken along line B1-B2 of FIG. 1. FIG. 2E is a sectional view taken along line B3-B4 of FIG. 1. FIG. 2F is a sectional view taken along line B5-B6 of FIG. 1.
- (22) As shown in FIG. 1, a sensor **110** according to the embodiment includes a first sensor part **10A**. The first sensor part **10A** includes an insulating member **41**, a first element part **21D**, and a second element part **22D**.
- (23) As shown in FIGS. 2A to 2F, the insulating member **41** may include, for example, a substrate **41s** and an insulating layer **41L** or the like. A part of the substrate **41s** may include a semiconductor layer such as a silicon substrate. For example, an insulating layer may be provided on the surface of the silicon substrate or the like, and various conductive layers may be provided on the insulating layer. The insulating layer **41L** may be provided on those conductive layers. In FIG. 1, the insulating layer **41L** is omitted.
- (24) As shown in FIG. 1, the first element part **21D** includes a first resistance element **21**, a first electrode **21U**, and a first other electrode **21V**. The first resistance element **21** includes a first resistance end portion **21e** and a first other resistance end portion **21f**. As shown in FIG. 2C, the first resistance element **21** is covered with the insulating member **41**.
- (25) As shown in FIG. 1, the first electrode **21U** is electrically connected to the first resistance end portion **21e**. As shown in FIG. 2A, the first electrode **21U** is not covered with the insulating member **41**.
- (26) As shown in FIG. 1, the first other electrode **21V** is electrically connected to the first other resistance end portion **21f**. As shown in FIG. 2B, the first other electrode **21V** is not covered with the insulating member **41**.
- (27) In this example, the first sensor part **10A** includes a first electrode wiring **21La** and a first other electrode wiring **21Lb**. The first electrode wiring **21La** electrically connects the first resistance end portion **21e** to the first electrode **21U**. In this example, the first electrode wiring **21La** is electrically connected to the first resistance end portion **21e** via a via **21va**. The first other electrode wiring **21Lb** electrically connects the first other resistance end portion **21f** to the first other electrode **21V**. In this example, the first other electrode wiring **21Lb** is electrically connected to the first other resistance end portion **21f** via a via **21vb**.
- (28) As shown in FIG. 1, the second element part **22D** includes a second resistance element **22**, a second electrode **22U**, and a second other electrode **22V**. The second resistance element **22** includes a second resistance end portion **22e** and a second other resistance end portion **22f**. As shown in FIG. 2F, the second resistance element **22** is covered with the insulating member **41**.
- (29) As shown in FIG. 1, the second electrode **22U** is electrically connected to the second resistance end portion **22e**. As shown in FIG. 2D, the second electrode **22U** is not covered with the insulating member **41**.

(30) As shown in FIG. 1, the second other electrode 22V is electrically connected to the second other resistance end portion 22f. As shown in FIG. 2E, the second other electrode 22V is not covered with the insulating member 41.

(31) In this example, the first sensor part 10A includes a second electrode wiring 22La and a second other electrode wiring 22Lb. The second electrode wiring 22La electrically connects the second resistance end portion 22e to the second electrode 22U. In this example, the second electrode wiring 22La is electrically connected to the second resistance end portion 22e via a via 22va. The second other electrode wiring 22Lb electrically connects the second other resistance end portion 22f to the second other electrode 22V. In this example, the second other electrode wiring 22Lb is electrically connected to the second other resistance end portion 22f via a via 22vb.

(32) At least any one of the first electrode wiring 21La, the first other electrode wiring 21Lb, the second electrode wiring 22La, or the second other electrode wiring 22Lb is a low resistance wiring. For example, at least any one of the first electrode wiring 21La, the first other electrode wiring 21Lb, the second electrode wiring 22La, or the second other electrode wiring 22Lb includes Al.

(33) As shown in FIG. 1, in the embodiment, a distance between the first electrode 21U and the first other electrode 21V is defined as a first distance L1. A distance between the second electrode 22U and the second other electrode 22V is defined as a second distance L2. The second distance L2 is shorter than the first distance L1.

(34) As shown in FIG. 1, for example, a parasitic resistance between the first electrode 21U and the first other electrode 21V is defined as a first parasitic resistance H1. A first electrical resistance Rp1 between the first electrode 21U and the first other electrode 21V corresponds to a resistance of a parallel circuit including the first parasitic resistance H1 and an electrical resistance R1 of the first resistance element 21.

(35) A parasitic resistance between the second electrode 22U and the second other electrode 22V is defined as a second parasitic resistance H2. A second electrical resistance Rp2 between the second electrode 22U and the second other electrode 22V corresponds to a resistance of a parallel circuit including the second parasitic resistance H2 and the electrical resistance R2 of the second resistance element 22.

(36) The second parasitic resistance H2 is affected by, for example, water adhering to the surface in the region between the second electrode 22U and the second other electrode 22V. For example, the second parasitic resistance H2 changes according to a change in humidity of the space around the first sensor part 10A. When the humidity is high, the parasitic resistance is low. When the humidity is low, the parasitic resistance is high. Therefore, the second electrical resistance Rp2 (parallel resistance) between the second electrode 22U and the second other electrode 22V changes according to the change in humidity of the space around the first sensor part 10A.

(37) On the other hand, the first electrical resistance Rp1 (parallel resistance) between the first electrode 21U and the first other electrode 21V does not substantially change according to a change in humidity. Alternatively, a first rate of change with respect to the change in humidity of the first electrical resistance Rp1 is lower than a second rate of change with respect to the change in humidity of the second electrical resistance Rp2. The first electrical resistance Rp1 (parallel resistance) is substantially the same as, for example, the electrical resistance R2 of the second resistance element 22.

(38) For example, by detecting such a difference between the first electrical resistance Rp1 and the second electrical resistance Rp2, humidity can be detected with high accuracy. According to the embodiment, it is possible to provide a sensor capable of improving the detection performance. For example, differential detection cancels temperature dependence of the electrical resistance R1 and the electrical resistance R2. For example, it is possible to provide a sensor whose characteristics change depending on changes in humidity.

(39) As shown in FIGS. 2A to 2F, the substrate 41s spreads along an X-Y plane. A direction from the substrate 41s to the first electrode 21U is along a Z-axis direction. The Z-axis direction is

perpendicular to an X-axis direction and a Y-axis direction. The Y-axis direction is perpendicular to the X-axis direction.

(40) As shown in FIG. 1, the sensor **110** may include a controller **70**. The controller **70** includes, for example, a first supply circuit **78a**, a second supply circuit **78b**, and a differential circuit **71**. The first supply circuit **78a** supplies a first current to the first resistance element **21** via the first electrode **21U** and the first other electrode **21V**. The second supply circuit **78b** supplies a second current to the second resistance element **22** via the second electrode **22U** and the second other electrode **22V**. The first supply circuit **78a** and the second supply circuit **78b** are, for example, constant current sources.

(41) The differential circuit **71** includes a first input part **71a**, a second input part **71b**, and an output part **71o**. A first voltage **v1** between the first electrode **21U** and the first other electrode **21V** is input to the first input part **71a**. A second voltage **v2** between the second electrode **22U** and the second other electrode **22V** is input to the second input part **71b**. The output part **71o** outputs a first signal **Sg1** according to a difference between the first voltage **v1** and the second voltage **v2**.

(42) The first signal **Sg1** includes information about humidity. For example, the humidity when the absolute value of the first signal **Sg1** is large is higher than the humidity when the absolute value of the first signal **Sg1** is small.

(43) For example, the electrical resistance **R1** of the first resistance element **21** may be substantially the same as the electrical resistance **R2** of the second resistance element **22**. For example, the electrical resistance **R1** of the first resistance element **21** is not less than 0.9 times and not more than 1.1 times the electrical resistance **R2** of the second resistance element **22**. Since these electrical resistances are substantially the same, the component corresponding to humidity becomes large in the first signal **Sg1**. Humidity is detected with higher accuracy.

(44) As shown in FIGS. 2A, 2B, 2D and 2E, the first electrode wiring **21La**, the first other electrode wiring **21Lb**, the second electrode wiring **22La** and the second other electrode wiring **22Lb** may be covered with the insulating member **41**.

(45) The first resistance element **21** may be formed of, for example, at least any one of a conductive layer **26a**, a conductive layer **27a**, a conductive layer **26b**, or a conductive layer **27b** (see FIGS. 2A and 2B). The second resistance element **22** may be formed of, for example, at least any one of a conductive layer **28a**, a conductive layer **29a**, a conductive layer **28b**, or a conductive layer **29b** (see FIGS. 2D and 2E). For example, at least any one of the first resistance element **21** and the second resistance element **22** includes TiN. It is easy to obtain a resistance of an appropriate value.

(46) In one example, at least any one of the first electrode **21U**, the first other electrode **21V**, the second electrode **22U** or the second other electrode **22V** includes at least one selected from the group consisting of Pd, Au, Pt and Al. For example, at least any one of the surface portions of the first electrode **21U**, the first other electrode **21V**, the second electrode **22U**, or the second other electrode **22V** includes at least one selected from the group consisting of Pd, Au, Pt, and Al. Corrosion and the like can be suppressed in these electrodes exposed to the outside. For example, stable characteristics can be easily obtained. These electrodes are, for example, electrode pads.

(47) The insulating member **41** includes, for example, silicon, and at least one selected from the group consisting of oxygen and nitrogen. The insulating member **41** includes, for example, at least any one of silicon nitride, silicon oxide or silicon oxynitride.

(48) In the following, an example of the experimental results conducted by the inventor will be described.

(49) In the experiment, a capacitive element with two electrodes, and two electrodes connected to those electrodes are provided on the substrate. The impedance between the two electrodes is measured by changing the humidity of the surrounding environment.

(50) FIGS. 3A and 3B are graphs illustrating experimental results.

(51) The horizontal axis of FIG. 3A is a real number component of impedance “Real Z”. The

vertical axis is an imaginary number component of impedance “Imag Z”. As shown in FIG. 3A, when the humidity is 30% RH, the real number component “Real Z” is substantially zero. In this case, the impedance between the two electrodes is only the capacitive component. On the other hand, when the humidity is 90% RH, the real number component “Real Z” is generated. In this case, the impedance between the two electrodes is the impedance of the parallel circuit of the capacitance component and the resistance component.

(52) The horizontal axis of FIG. 3B is a frequency  $f_{m1}$  in the measurement of impedance. The vertical axis is a parallel resistance component  $R_{pr}$ . When the humidity is 30% RH, the parallel resistance component  $R_{pr}$  is high. On the other hand, when the humidity is 90% RH, the parallel resistance component  $R_{pr}$  decreases.

(53) In this way, the resistance component between the two electrodes changes depending on humidity. Humidity can be detected by utilizing this characteristic.

(54) In the following, an example of the simulation result regarding the electrical resistance  $R_2$  of the second resistance element 22 and the second parasitic resistance  $H_2$  between the second electrode 22U and the second other electrode 22V will be described.

(55) FIGS. 4A and 4B are schematic views illustrating simulation results.

(56) FIG. 4A illustrates a simulation model. FIG. 4B illustrates the simulation results. As shown in FIG. 4A, in the simulation model, the electrical resistance  $R_2$  of the second resistance element 22 and the second parasitic resistance  $H_2$  between the second electrode 22U and the second other electrode 22V are connected in parallel. A combined resistance of the parallel circuit is defined as a resistance  $R_x$ .

(57) The horizontal axis of FIG. 4B is the second parasitic resistance  $H_2$  between the second electrode 22U and the second other electrode 22V. The vertical axis is a resistance change  $\Delta R$ . The resistance change  $\Delta R$  is a difference between the electrical resistance  $R_2$  and the resistance  $R_x$ . As shown in FIG. 4B, when the electrical resistance  $R_2$  of the second resistance element 22 is as low as  $100\Omega$ , the resistance change  $\Delta R$  is substantially 0 even if the second parasitic resistance  $H_2$  changes. When the electrical resistance  $R_2$  of the second resistance element 22 becomes as high as  $2\text{ k}\Omega$  or  $4\text{ k}\Omega$ , the resistance change  $\Delta R$  changes with a change of the second parasitic resistance  $H_2$ . As described above, the second parasitic resistance  $H_2$  changes depending on humidity.

(58) In the embodiment, the electrical resistance  $R_2$  of the second resistance element 22 is preferably not less than  $1\text{ k}\Omega$  and not more than  $20\text{ k}\Omega$ , for example. This makes it easy to detect changes in humidity appropriately and with high accuracy. As described above, it is preferable that the electrical resistance  $R_1$  of the first resistance element 21 is substantially the same as the electrical resistance  $R_2$ . Therefore, the electrical resistance  $R_1$  of the first resistance element 21 is preferably not less than  $1\text{ k}\Omega$  not more than and  $20\text{ k}\Omega$ .

(59) As described below, the second electrode 22U and the second other electrode 22V may be comb-shaped electrodes.

(60) FIGS. 5A and 5B are schematic plan views illustrating the sensor according to the first embodiment.

(61) As shown in FIGS. 5A and 5B, the second electrode 22U includes a second electrode base 22Ap and multiple second electrode protrusions 22Aq. The multiple second electrode protrusions 22Aq are electrically connected to the second electrode base 22Ap. At least a part of the second other electrode 22V is between one of the multiple second electrode protrusions 22Aq and another one of the multiple second electrode protrusions 22Aq.

(62) In these examples, the second other electrode 22V includes a second other electrode base 22Bp and a multiple second other electrode protrusions 22Bq. The multiple second other electrode protrusions 22Bq are electrically connected to the second other electrode base 22Bp. One of the multiple second electrode protrusions 22Aq is between one of the multiple second other electrode protrusions 22Bq and another one of the multiple second other electrode protrusions 22Bq.

(63) Due to the comb-shaped electrode, the second parasitic resistance  $H_2$  between the second

electrode **22U** and the second other electrode **22V** is likely to change with higher sensitivity to changes in humidity. Due to the comb-shaped electrode, the second electric resistance **Rp2** (parallel resistance) between the second electrode **22U** and the second other electrode **22V** is likely to change with higher sensitivity to changes in humidity.

## Second Embodiment

(64) A sensor according to a second embodiment also includes the first sensor part **10A** (see FIG. **1**). Also in the second embodiment, the first sensor part **10A** includes the insulating member **41**, the first element part **21D**, and the second element part **22D** (see FIG. **1**). Also in the second embodiment, the first element part **21D** includes the first resistance element **21**, the first electrode **21U**, and the first other electrode **21V** (see FIG. **1**). Also in the second embodiment, the second element part **22D** includes the second resistance element **22**, the second electrode **22U**, and the second other electrode **22V** (see FIG. **1**). In the second embodiment, characteristics in the region between the second electrode **22U** and the second other electrode **22V** are different from characteristics in a region between the first electrode **21U** and the first other electrode **21V**. In the following, an example of the characteristics between these electrodes will be described with respect to the second embodiment.

(65) FIGS. **6A** and **6B** are schematic cross-sectional views illustrating a sensor according to a second embodiment.

(66) As shown in FIG. **6A**, in a sensor **120**, the insulating member **41** includes a first insulating region **45a** between the first electrode **21U** and the first other electrode **21V**. As shown in FIG. **6B**, the insulating member **41** includes a second insulating region **45b** between the second electrode **22U** and the second other electrode **22V**.

(67) The insulating member **41** has a first material of the first insulating region **45a** and a first contact angle of the first insulating region **45a** with respect to water. The insulating member **41** has at least any one of a second material of the second insulating region **45b** and a second contact angle of the second insulating region **45b** with respect to water. The second material is different from the first material. The second contact angle is smaller than the first contact angle. Except for this, at least a part of the configuration of the sensor **120** may be the same as the configuration of the sensor **110**.

(68) For example, the first insulating region **45a** includes the first material. The second insulating region **45b** includes the second material. The second material is different from the first material. For example, the contact angle of the second insulating region **45b** with water is smaller than the contact angle of the first insulating region **45a** with water.

(69) As described above, the materials and surface characteristics of the first insulating region **45a** and the second insulating region **45b** are different from each other. This causes the electrical resistance between these electrodes to change with humidity. Humidity can be detected by detecting a change in a value (e.g., voltage) corresponding to the electrical resistance between these electrodes.

(70) In the second embodiment, the first distance **L1** between the first electrode **21U** and the first other electrode **21V** may be different from or the same as the second distance **L2** between the second electrode **22U** and the second other electrode **22V** (see FIG. **1**).

(71) In the sensor **120**, for example, the contact angle of the second material with water is smaller than the contact angle of the first material with water. In the first material, for example, hydrophobicity (for example, water repellency) is high. In the second material, for example, the hydrophilicity is high. The first material may include at least any one of fluorine or an alkyl group. The alkyl group may include, for example, a methyl group. The first material may include, for example, polydimethylsiloxane (PDMS) and the like.

(72) A difference in characteristics between the first insulating region **45a** and the second insulating region **45b** may be provided due to a difference in surface treatment. For example, the hydrophilicity is increased by irradiation with electromagnetic waves such as ultraviolet rays. For

example, treatment with particles or the like increases hydrophilicity. For example, the contact angle can be changed by treatment with a surface modifier (for example, a silane coupling agent). This makes it possible to obtain a difference in the change in parasitic resistance according to the change in humidity.

### Third Embodiment

(73) FIG. 7 is a schematic plan view illustrating a sensor according to a third embodiment.

(74) FIG. 8 is a schematic cross-sectional view illustrating a part of the sensor according to the third embodiment.

(75) As shown in FIG. 7, a sensor **130** according to the embodiment further includes a second sensor part **10B** in addition to the first sensor part **10A**. In the sensor **130**, the configuration of the first sensor part **10A** may be the same as the configuration of the first sensor part **10A** in the sensor **110** or the sensor **120**.

(76) As shown in FIG. 8, the second sensor part **10B** includes a base body **41B**, a fixed electrode **35**, a support portion **33**, and a movable electrode **31**. The fixed electrode **35** is fixed to the base body **41B**. The support portion **33** is fixed to the base body **41B**. The movable electrode **31** is supported by the support portion **33**. A first void **g1** is provided between the fixed electrode **35** and the movable electrode **31**.

(77) The base body **41B** may be continuous with the substrate **41s**. The base body **41B** may be provided separately from the substrate **41s**.

(78) In this example, the second sensor part **10B** further includes a connection portion **32**. The connection portion **32** is supported by the support portion **33**. The connection portion **32** supports the movable electrode **31**.

(79) A first value corresponding to the capacitance between the fixed electrode **35** and the movable electrode **31** changes depending on a substance to be detected included in a space around the second sensor part **10B**. The substance to be detected is, for example, hydrogen.

(80) For example, the connection portion **32** can be deformed according to a concentration of the substance to be detected included in the space. For example, a substance adsorption layer **32L** is provided on the connection portion **32**. The substance adsorption layer **32L** expands according to the degree of adsorption of the substance. The connection portion **32** is deformed by the deformation of the substance adsorption layer **32L**. A distance between the movable electrode **31** and the fixed electrode **35** changes according to the deformation of the connection portion **32**. By measuring the capacitance according to the change in distance, the concentration of the substance to be detected can be detected.

(81) As shown in FIG. 8, the connection portion **32** may include a heater **32H**. The connection portion **32** is locally heated by the heater **32H**. As a result, the water adsorbed on the substance adsorption layer **32L** (for example, the gas sensitive film) can be released. For example, oxygen adsorbed on the substance adsorption layer **32L** can be efficiently reduced in a reducing gas atmosphere such as hydrogen.

(82) The movable electrode **31** illustrated in FIG. 8 may be the same layer as any one of multiple of conductive layers (conductive layer **26a**, conductive layer **27a**, conductive layer **26b**, conductive layer **27b**, conductive layer **28a**, conductive layer **29a**, conductive layer **28b**, and conductive layer **29b**, and the like) described with respect to FIGS. 2A to 2F. The heater **32H** may be the same layer as any one of the multiple conductive layers described with respect to FIGS. 2A to 2F.

(83) In the sensor **130**, the controller **70** further includes a capacitance detector **72**. The capacitance detector **72** can detect the capacitance between the fixed electrode **35** and the movable electrode **31**. For example, the capacitance detector **72** can output the first value corresponding to the capacitance between the fixed electrode **35** and the movable electrode **31**. The controller **70** can output a second value obtained by correcting the first value using the first signal **Sg1**.

(84) As shown in FIGS. 7 and 8, in the second sensor part **10B**, a movable electrode terminal **38a** and a fixed electrode terminal **38b** are provided. The movable electrode terminal **38a** is electrically

connected to the movable electrode **31**. The fixed electrode terminal **38b** is electrically connected to the fixed electrode **35**. For example, the movable electrode terminal **38a** is electrically connected to the second other electrode **22V** via a wiring **31La** and a via **25v**. The fixed electrode terminal **38b** is electrically connected to the second electrode **22U** by a wiring **31Lb**.

(85) The capacitance between the fixed electrode **35** and the movable electrode **31** is detected by the capacitance detector **72** via the second electrode **22U** and the second other electrode **22V**. In order to reduce the size of the sensor **130**, the second distance **L2** between the second electrode **22U** and the second other electrode **22V** may be set short. In this case, in the measurement of the capacitance between the fixed electrode **35** and the movable electrode **31**, the influence of the second parasitic resistance **H2** between the second electrode **22U** and the second other electrode **22V** occurs. In the embodiment, the first value of the capacitance measurement is corrected by using the first signal **Sg1** corresponding to the change of the second parasitic resistance **H2**. High accuracy can be obtained in the corrected second value.

(86) As shown in FIG. 7, in this example, a first switch **SW1** is provided in a current path (second other electrode wiring **22Lb**) between the second other electrode **22V** and the second resistance element **22**. As shown in FIG. 7, for example, the first other electrode **21V** is set to the fixed potential **GND**. The fixed potential **GND** is, for example, a ground potential. In this example, a second switch **SW2** is provided in a current path between the first other electrode **21V** and the second other electrode **22V**. Multiple operations may be performed on the sensor **130** by operating these switches.

(87) FIG. 9 is a schematic plan view illustrating one of multiple operations in the sensor according to the third embodiment.

(88) FIG. 9 corresponds to a first operation **OP1**. The first operation **OP1** corresponds to, for example, a resistance measurement mode. As shown in FIG. 9, in the first operation **OP1**, the first switch **SW1** and the second switch **SW2** are in a conductive state. At this time, the second resistance element **22** is connected to the second other electrode **22V**, and the second other electrode **22V** is set to the fixed potential **GND**. In this first operation **OP1**, for example, the operation described with respect to the sensor **110** is performed. For example, the differential circuit **71** outputs the first signal **Sg1** according to the difference between the first voltage **v1** related to the first resistance element **21** and the second voltage **v2** related to the second resistance element **22**. That is, humidity is detected.

(89) FIG. 10 is a schematic plan view illustrating one of multiple operations in the sensor according to the third embodiment.

(90) FIG. 10 corresponds to a second operation **OP2**. The second operation **OP2** corresponds to a capacitance measurement mode. As shown in FIG. 10, in the second operation **OP2**, the first switch **SW1** and the second switch **SW2** are in a non-conducting state. At this time, the second resistance element **22** is insulated from the second other electrode **22V**. The second other electrode **22V** is insulated from the fixed potential **GND**. In the second operation **OP2**, a change in the capacitance in the second sensor part **10B** is detected. For example, the change in the capacitance is detected by the capacitance detector **72**. The change in the capacitance is the change in the capacitance between the fixed electrode **35** and the movable electrode **31**. The change in the capacitance changes depending on the substance to be detected (for example, hydrogen). In the second operation **OP2**, the first value corresponding to the concentration of the substance to be detected is derived, and the first value is corrected based on the first signal **Sg1**. The corrected second value is output from, for example, the controller **70**.

(91) FIG. 11 is a schematic plan view illustrating the sensor according to the third embodiment.

(92) As shown in FIG. 11, in the sensor **130**, the controller **70** may include a control circuit **75**. The control circuit **75** supplies a first control signal **Sc1** to the first switch **SW1**. The opening/closing operation of the first switch **SW1** is controlled by the first control signal **Sc1**. The control circuit **75** supplies a second control signal **Sc2** to the second switch **SW2**. The opening/closing operation of

the second switch SW2 is controlled by the second control signal Sc2. The control circuit 75 may supply a third control signal Sc3 to the capacitance detector 72. The operation of the capacitance detector 72 is controlled by the third control signal Sc3.

(93) As shown in FIG. 11, the controller 70 may include the differential circuit 71. The first voltage v1 and the second voltage v2 are input to the differential circuit 71. The output of the differential circuit 71 may be supplied to the control circuit 75. The differential circuit 71 may be included in, for example, an ADC (Analog-to-digital Converter). In this example, the controller 70 includes the capacitance detector 72, the first switch SW1 and the second switch SW2.

(94) In the embodiment, the first value before correction (the value corresponding to the capacitance between the fixed electrode 35 and the movable electrode 31) greatly increases as humidity increases. On the other hand, the corrected second value does not substantially change even if humidity changes. By the correction, the substance to be detected can be detected with high accuracy by suppressing the influence of humidity.

(95) FIG. 12 is a block diagram illustrating a sensor according to the third embodiment.

(96) As shown in FIG. 12, in a sensor 131 according to the embodiment, the first electrical resistance Rp1 and the second electrical resistance Rp2 are provided in the first sensor part 10A. In the second sensor part 10B, a first capacitance C1 is provided between the fixed electrode 35 and the movable electrode 31. The heater (heater 32H: see FIG. 8) is provided in the second sensor part 10B.

(97) In the example shown in FIG. 12, for example, the controller 70 is provided on a module board 76S. As shown in FIG. 12, the controller 70 includes an ADC. The ADC includes, for example, the differential circuit 71. The first voltage v1 corresponding to the first electrical resistance Rp1 and the second voltage v2 corresponding to the second electrical resistance Rp2 are input to the ADC (differential circuit 71). For example, humidity is detected by detecting the difference between the first voltage v1 and the second voltage v2. For example, humidity is detected by detecting the difference between the first electrical resistance Rp1 and the second electrical resistance Rp2.

(98) For example, a voltage v0 of a reference resistance “R\_ref” may be input to the ADC. A temperature coefficient of the reference resistance “R\_ref” is low. For example, the temperature is detected by detecting a difference between the voltage v0 and the first voltage v1. For example, the temperature is detected by detecting a difference between the reference resistance “R\_ref” and the first electrical resistance Rp1. The reference resistance “R\_ref” and the first electrical resistance Rp1 correspond to a temperature sensor, for example. The detection result may be corrected by the temperature detected by the temperature sensor.

(99) The controller 70 includes, for example, an MCU (Micro Computer Unit). The first signal Sg1 output from the ADC is supplied to the MCU. The MCU may correspond to, for example, the control circuit 75.

(100) The controller 70 includes, for example, a CDC (Capacitance-to-digital Converter). The CDC converts the change in the first capacitance C1 into a digital signal. The CDC corresponds to a part of the capacitance detector 72.

(101) The signal (first value) output from the CDC is supplied to the MCU. In the MCU, the first value is corrected and the second value is output based on the first signal Sg1. The first signal Sg1 may include information on humidity and information on temperature. The first value may be corrected and the second value may be output based on at least any one of temperature or humidity.

(102) As shown in FIG. 12, the controller 70 may include a power supply circuit 76a. The controller 70 may include a communication circuit 76b. The communication circuit 76b may include a wireless transmission/reception circuit. The controller 70 may include a booster circuit 76c. A current may be supplied to the heater from the booster circuit 76c. In the embodiment, the booster circuit 76c may be omitted. For example, a current may be supplied to the heater from the output unit of the MCU.

(103) The following sequence of operations may be performed on the sensor **131**.

(104) For example, in the first step, the temperature is detected by measuring the difference between the first voltage **v1** and the voltage **v0** of the reference resistance “R\_ref”. In the second step, the parasitic resistance is detected based on the difference between the first voltage **v1** and the second voltage **v2**. The first signal **Sg1** is generated. In the third step, a current is supplied to the heater **32H**, and the temperature of the connection portion **32** is raised.

(105) In the embodiment, for example, moisture is not substantially adsorbed on a portion other than the surface of the sensor. For example, only fluctuations due to moisture adsorption between pads are detected. For example, an analog wiring included in the sensor may be sandwiched between the ground layers (metal layer). As a result, moisture adsorption can be suppressed around the analog wiring.

(106) FIG. **13** is a schematic cross-sectional view illustrating a sensor according to the embodiment.

(107) As shown in FIG. **13**, a sensor **140** according to the embodiment includes a container **61** and a circuit board **62** in addition to the first sensor part **10A**. At least a part of the controller **70** is sealed by the container **61** and the circuit board **62**. For example, at least any one of the CDC or ADC included in the controller **70** is sealed by the container **61** and the circuit board **62**. Also in this case, the first electrode **21U**, the first other electrode **21V**, the second electrode **22U**, and the second other electrode **22V** are not covered with the container **61** and the circuit board **62**.

(108) The circuit board **62** may include, for example, a first ground layer **62a**, an analog signal layer **62b**, a second ground layer **62c**, a digital signal layer **62d**, and the like. These layers may be provided in the insulating member **41**. The analog signal layer **62b**, the second ground layer **62c**, and the digital signal layer **62d** are provided between the first ground layer **62a** and the container **61**. Noise is more suppressed.

(109) According to the embodiment, it is possible to provide a sensor capable of improving the detection performance.

(110) The embodiments may include the following configurations (e.g., technological proposals).

(111) Configuration 1

(112) A sensor, comprising: a first sensor part including an insulating member, a first element part including a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member, and a second element part including a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member, a second distance between the second electrode and the second other electrode being shorter than a distance between the first electrode and the first other electrode.

Configuration 2

(113) A sensor, comprising: a first sensor part including an insulating member, a first element part including a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member, and a second element part including a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically

connected to the second other resistance end portion, and not covered with the insulating member, the insulating member including a first insulating region between the first electrode and the first other electrode, and a second insulating region between the second electrode and the second other electrode, the insulating member including a first material of the first insulating region, and a first contact angle of the first insulating region with respect to water, and the insulating member including at least any one of a second material of the second insulating region, the second material being different from the first material, or a second contact angle of the second insulating region with respect to water, the second contact angle being smaller than the first contact angle.

#### Configuration 3

(114) The sensor according to Configuration 1, wherein a contact angle of the second material with respect to water is smaller than a contact angle of the first material with respect to water.

#### Configuration 4

(115) The sensor according to Configuration 3, wherein the first material includes at least any one of fluorine or an alkyl group.

#### Configuration 5

(116) The sensor according to any one of Configurations 1 to 4, wherein the second electrode includes a second electrode base, and a plurality of second electrode protrusions electrically connected to the second electrode base, and at least a part of the second other electrode is between one of the plurality of second electrode protrusions and an other one of the plurality of second electrode protrusions.

#### Configuration 6

(117) The sensor according to Configuration 5, wherein the second other electrode includes a second other electrode base, and a plurality of second other electrode protrusions electrically connected to the second other electrode base, and the one of the plurality of second electrode protrusions is between one of the plurality of second other electrode protrusions and an other one of the plurality of second other electrode protrusions.

#### Configuration 7

(118) The sensor according to any one of Configurations 1 to 6, wherein an electrical resistance of the first resistance element is not less than 0.9 times and not more than 1.1 times an electrical resistance of the second resistance element.

#### Configuration 8

(119) The sensor according to any one of Configurations 1 to 6, wherein an electrical resistance of the second resistance element is not less than 1 k $\Omega$  and not more than 20 k $\Omega$ .

#### Configuration 9

(120) The sensor according to any one of Configurations 1 to 8, wherein at least any one of the first electrode, the first other electrode, the second electrode, or the second other electrode includes at least one selected from the group consisting of Pd, Au, Pt, and Al.

#### Configuration 10

(121) The sensor according to any one of Configurations 1 to 9, wherein at least any one of the first resistance element or the second resistance element includes TiN.

#### Configuration 11

(122) The sensor according to any one of Configurations 1 to 10, wherein the first sensor part further includes a first electrode wiring electrically connecting the first resistance end portion to the first electrode, a first other electrode wiring electrically connecting the first other resistance end portion to the first other electrode, a second electrode wiring electrically connecting the second resistance end portion to the second electrode, and a second other electrode wiring electrically connecting the second other resistance end portion to the second other electrode, and at least any one of the first electrode wiring, the first other electrode wiring, the second electrode wiring, or the second other electrode wiring includes Al.

#### Configuration 12

(123) The sensor according to any one of Configurations 1 to 11, wherein the insulating member includes silicon and at least one selected from the group consisting of oxygen and nitrogen.

#### Configuration 13

(124) The sensor according to any one of Configurations 1 to 12, wherein a second electrical resistance between the second electrode and the second other electrode changes depending on a change of humidity in a space around the first sensor part, and a first electrical resistance between the first electrode and the first other electrode does not change depending on the change of the humidity, or a first change rate of the first electrical resistance to the change of the humidity is lower than a second change rate of the second electrical resistance to the change of the humidity.

#### Configuration 14

(125) The sensor according to Configuration 13, further comprising: a controller, the controller including a first supply circuit supplying a first current to the first resistance element via the first electrode and the first other electrode, a second supply circuit supplying a second current to the second resistance element via the second electrode and the second other electrode, and a differential circuit, the differential circuit including a first input part to which a first voltage between the first electrode and the first other electrode is input, a second input part to which a second voltage between the second electrode and the second other electrode is input, and an output part outputting a first signal corresponding to a difference between the first voltage and the second voltage.

#### Configuration 15

(126) The sensor according to Configuration 14, further comprising: a second sensor part, the second sensor part including a base body, a fixed electrode fixed to the base body, a support portion fixed to the base body a movable electrode supported by the support portion, a first void being provided between the fixed electrode and the movable electrode, the controller further including a capacitance detector, the capacitance detector outputting a first value corresponding to a capacitance between the fixed electrode and the movable electrode, and the controller being configured to output a second value obtained by correcting the first value using the first signal.

#### Configuration 16

(127) The sensor according to Configuration 15, wherein the first value changes depending on substance to be detected included in a space around the second sensor part.

#### Configuration 17

(128) The sensor according to Configuration 16, wherein a first distance between the fixed electrode and the movable electrode changes depending on a concentration of the substance in the space.

#### Configuration 18

(129) The sensor according to Configuration 17, wherein the second sensor part further includes a connection portion supported by the support portion and supporting the movable electrode, and the connection portion being deformable depending on the concentration of the substance.

#### Configuration 19

(130) The sensor according to any one of Configurations 1 to 18, further comprising: a container; and a circuit board, at least a part of the controller being sealed by the container and the circuit board, and the first electrode, the first other electrode, the second electrode, and the second other electrode being not covered with the container and the circuit board.

(131) In the specification, “electrically connected” includes a case of being connected via other conductive member or the like other than a case of being connected by direct contact.

(132) In the specification of the application, “perpendicular” and “parallel” refer to not only strictly perpendicular and strictly parallel but also include, for example, the fluctuation due to manufacturing processes, etc. It is sufficient to be substantially perpendicular and substantially parallel.

(133) Hereinabove, exemplary embodiments of the invention are described with reference to

specific examples. However, the embodiments of the invention are not limited to these specific examples. For example, one skilled in the art may similarly practice the invention by appropriately selecting specific configurations of components included in sensors such as sensor parts, resistance elements, electrodes, controllers, etc., from known art. Such practice is included in the scope of the invention to the extent that similar effects thereto are obtained.

(134) Further, any two or more components of the specific examples may be combined within the extent of technical feasibility and are included in the scope of the invention to the extent that the purport of the invention is included.

(135) Moreover, all sensors practicable by an appropriate design modification by one skilled in the art based on the sensors described above as embodiments of the invention also are within the scope of the invention to the extent that the spirit of the invention is included.

(136) Various other variations and modifications can be conceived by those skilled in the art within the spirit of the invention, and it is understood that such variations and modifications are also encompassed within the scope of the invention.

(137) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention.

## Claims

1. A sensor, comprising: a first sensor part including an insulating member, a first element part including a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member, and a second element part including a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member, a second distance between the second electrode and the second other electrode being shorter than a distance between the first electrode and the first other electrode, wherein the first sensor part further includes a first electrode wiring electrically connecting the first resistance end portion to the first electrode, a first other electrode wiring electrically connecting the first other resistance end portion to the first other electrode, a second electrode wiring electrically connecting the second resistance end portion to the second electrode, and a second other electrode wiring electrically connecting the second other resistance end portion to the second other electrode, and at least any one of the first electrode wiring, the first other electrode wiring, the second electrode wiring, or the second other electrode wiring includes Al.

2. The sensor according to claim 1, wherein the second electrode includes a second electrode base, and a plurality of second electrode protrusions electrically connected to the second electrode base, and at least a part of the second other electrode is between one of the plurality of second electrode protrusions and an other one of the plurality of second electrode protrusions.

3. The sensor according to claim 2, wherein the second other electrode includes a second other electrode base, and a plurality of second other electrode protrusions electrically connected to the second other electrode base, and the one of the plurality of second electrode protrusions is between

one of the plurality of second other electrode protrusions and an other one of the plurality of second other electrode protrusions.

4. The sensor according to claim 1, wherein an electrical resistance of the first resistance element is not less than 0.9 times and not more than 1.1 times an electrical resistance of the second resistance element.

5. The sensor according to claim 1, wherein an electrical resistance of the second resistance element is not less than 1 k $\Omega$  and not more than 20 k $\Omega$ .

6. The sensor according to claim 1, wherein at least any one of the first electrode, the first other electrode, the second electrode, or the second other electrode includes at least one selected from the group consisting of Pd, Au, Pt, and Al.

7. The sensor according to claim 1, wherein at least any one of the first resistance element or the second resistance element includes TiN.

8. The sensor according to claim 1, wherein the insulating member includes silicon and at least one selected from the group consisting of oxygen and nitrogen.

9. The sensor according to claim 1, wherein a second electrical resistance between the second electrode and the second other electrode changes depending on a change of humidity in a space around the first sensor part, and a first electrical resistance between the first electrode and the first other electrode does not change depending on the change of the humidity, or a first change rate of the first electrical resistance to the change of the humidity is lower than a second change rate of the second electrical resistance to the change of the humidity.

10. The sensor according to claim 9, further comprising: a controller, the controller including a first supply circuit supplying a first current to the first resistance element via the first electrode and the first other electrode, a second supply circuit supplying a second current to the second resistance element via the second electrode and the second other electrode, and a differential circuit, the differential circuit including a first input part to which a first voltage between the first electrode and the first other electrode is input, a second input part to which a second voltage between the second electrode and the second other electrode is input, and an output part outputting a first signal corresponding to a difference between the first voltage and the second voltage.

11. The sensor according to claim 10, further comprising: a second sensor part, the second sensor part including a base body, a fixed electrode fixed to the base body, a support portion fixed to the base body a movable electrode supported by the support portion, a first void being provided between the fixed electrode and the movable electrode, the controller further including a capacitance detector, the capacitance detector outputting a first value corresponding to a capacitance between the fixed electrode and the movable electrode, and the controller being configured to output a second value obtained by correcting the first value using the first signal.

12. The sensor according to claim 11, wherein the first value changes depending on substance to be detected included in a space around the second sensor part.

13. The sensor according to claim 12, wherein a first distance between the fixed electrode and the movable electrode changes depending on a concentration of the substance in the space.

14. The sensor according to claim 13, wherein the second sensor part further includes a connection portion supported by the support portion and supporting the movable electrode, and the connection portion being deformable depending on the concentration of the substance.

15. The sensor according to claim 10, further comprising: a container; and a circuit board, at least a part of the controller being sealed by the container and the circuit board, and the first electrode, the first other electrode, the second electrode, and the second other electrode being not covered with the container and the circuit board.

16. A sensor, comprising: a first sensor part including an insulating member, a first element part including a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a

first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member, and a second element part including a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member, the insulating member including a first insulating region between the first electrode and the first other electrode, and a second insulating region between the second electrode and the second other electrode, the insulating member including a first material of the first insulating region, and a first contact angle of the first insulating region with respect to water, and the insulating member including at least any one of a second material of the second insulating region, the second material being different from the first material, or a second contact angle of the second insulating region with respect to water, the second contact angle being smaller than the first contact angle, wherein the first sensor part further includes a first electrode wiring electrically connecting the first resistance end portion to the first electrode, a first other electrode wiring electrically connecting the first other resistance end portion to the first other electrode, a second electrode wiring electrically connecting the second resistance end portion to the second electrode, and a second other electrode wiring electrically connecting the second other resistance end portion to the second other electrode, and at least any one of the first electrode wiring, the first other electrode wiring, the second electrode wiring, or the second other electrode wiring includes Al.

17. The sensor according to claim 16, wherein a contact angle of the second material with respect to water is smaller than a contact angle of the first material with respect to water.

18. The sensor according to claim 17, wherein the first material includes at least any one of fluorine or an alkyl group.

19. A sensor, comprising: a first sensor part including an insulating member, a first element part including a first resistance element including a first resistance end portion and a first other resistance end portion, and covered with the insulating member, a first electrode electrically connected to the first resistance end portion, and not covered with the insulating member, and a first other electrode electrically connected to the first other resistance end portion, and not covered with the insulating member, and a second element part including a second resistance element including a second resistance end portion and a second other resistance end portion, and covered with the insulating member, a second electrode electrically connected to the second resistance end portion, and not covered with the insulating member, and a second other electrode electrically connected to the second other resistance end portion, and not covered with the insulating member, a second distance between the second electrode and the second other electrode being shorter than a distance between the first electrode and the first other electrode, wherein a second electrical resistance between the second electrode and the second other electrode changes depending on a change of humidity in a space around the first sensor part, and a first electrical resistance between the first electrode and the first other electrode does not change depending on the change of the humidity, or a first change rate of the first electrical resistance to the change of the humidity is lower than a second change rate of the second electrical resistance to the change of the humidity.

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